

KAIYUAN TAN

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BIOGRAPHY

I am currently a **Ph.D. student** at the Department of Computer Science, **Vanderbilt University**, where I am dedicated to developing safe and robust intelligent robotics systems. My research encompasses physics-informed dynamics learning, learning-enabled control, and planning, with a focus on enhancing the reliability and performance of robotic platforms in complex environments.

Research Interests: Robotics, Physics-informed Machine Learning, Adversarial Attack, Learning-based Control

EDUCATION

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|---|---------------------|
| Ph.D. in Computer Science
Vanderbilt University, TN, USA
Advisor: Dr. Thomas Beckers
Research Focus: “Physics-informed Machine Learning” | Jan 2024 - Current |
| M.S. in Electrical Engineering
Washington University in St. Louis, St. Louis, MO, USA
Advisor: Dr. Yiannis Kantaros
Thesis: “Targeted Adversarial Attack Generation and Detection in Autonomous Driving System” | Aug 2021 - May 2023 |
| B.Eng. in Information Engineering
Sun Yat-Sen University, Guangzhou, Guangdong, China
Advisor: Dr. Lei Sun
Thesis: “Estimation of mixed noise parameters depending on hyperspectral images signals” | Aug 2017 - May 2021 |

WORK EXPERIENCE

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| Research Associate Student (Robotics & Learning-Enabled Control)
Vanderbilt University, Nashville, TN, USA
Advisor: Dr. Thomas Beckers | Aug 2023 - Current |
| Research Associate Intern (Robotics & Cyber-Physical System Security)
Washington University in St. Louis, St. Louis, MO, USA
Advisor: Dr. Ning Zhang | May 2023 - July 2023 |
| Research Associate Student (Adversarial Attack & Learning-Enabled Control)
Washington University in St. Louis, St. Louis, MO, USA
Advisor: Dr. Yiannis Kantaros | June 2022 - May 2023 |

PUBLICATIONS

- [1] J. Wang, Jiaming Tong, **K. Tan**, and Y. Kantaros, “Large Language Model Based Robot Planning with Safety-Guaranteed Temporal Logic Task Specification.” *ACM Transactions on Cyber-Physical Systems (TCPS)*, 2025.
- [2] **K. Tan**, P. Li, J. Wang, and T. Beckers, “PnP-PIML: Physics-informed Learning of Outlier Dynamics using Uncertainty Quantified Port-Hamiltonian Models.” *IEEE Conference on Robotics and Automation (ICRA)*, 2025.
- [3] P. Li, **K. Tan**, and T. Beckers, “NAPI-MPC: Neural Accelerated Physics-Informed MPC for Nonlinear PDE Systems.” *6th Annual Learning for Dynamics & Control Conference at UMich (L4DC)*, 2025.
- [4] J. Wang, **K. Tan**, Z. Sun, and Y. Kantaros, “Mission-driven Exploration for Accelerated Deep Reinforcement Learning with Temporal Logic Task Specifications.” *6th Annual Learning for Dynamics & Control Conference*

at UMich (**L4DC**), 2025.

- [5] **K. Tan**, P. Li, and T. Beckers, “Physics-Constrained Learning of PDE Systems with Uncertainty Quantified Port-Hamiltonian Models.” 6th Annual Learning for Dynamics & Control Conference at Oxford (**L4DC**), 2024.
- [6] P. Li, **K. Tan**, and T. Beckers, “PyGpPHs: A Python Package for Bayesian Modeling of Port-Hamiltonian Systems.” (**IFAC-PapersOnLine**), 2024.
- [7] **K. Tan**, J. Wang, and Y. Kantaros, “Targeted Adversarial Attacks against Neural Network Trajectory Predictors.” 5th Annual Learning for Dynamics & Control Conference at UPenn (**L4DC**), 2023.
- [8] **K. Tan**, Z. Wang, Z. Liu, and L. Sun, “Research on Full-Space Spectrum-Sharing Strategy for Massive MIMO Cognitive Radio Systems” The 2021 IEEE International Conference on Consumer Electronics and Computer Engineering (ICCECE), (pp. 639-644). IEEE.

TECHNICAL SKILLS

Programming Languages Python, Matlab, C/C++
Deep Learning Frameworks Pytorch

AWARDS

S Award, US Mathematical Contest in Modeling, 2018-2019.